

Placement Readiness Digital Twin & Analytics Dashboard

An AI-Powered Enterprise Decision-Support & Student Career Twin Platform.
Integrates Machine Learning Student Outcome Simulation with Macro-Level Branch Trends, Skills Intelligence, Recruiter Performance, and Salary CTC Analytics.

Author / System:	Placement Intelligence & Digital Twin Systems
Version:	2.0.0 (Analytics & Model Integrated)
Production GitHub:	https://github.com/sashi299/placement-predictor
Base Repository:	https://github.com/Vivek-Matrix/placement-readiness-digital-twin
Core Frameworks:	Python 3.10+, Streamlit, Scikit-Learn, Plotly, Pandas, Joblib
Automated Testing:	Playwright MCP End-to-End Test Engine
Publication Date:	September 2026
Build Status:	PRODUCTION-VERIFIED & DEPLOYMENT-READY

1. Executive Summary & Architecture

The **Placement Readiness Digital Twin & Analytics Portal** is an integrated, full-stack data science platform designed to solve two critical challenges in academic career management: (1) **Micro-Level Simulation**: Predicting individual student readiness and company fit using a multi-dimensional digital twin powered by Machine Learning, and (2) **Macro-Level Analytics**: Providing institutional stakeholders with deep intelligence into branch-wise trends, skill proficiencies, recruiter hiring volume, and compensation (CTC) statistics.

System Architecture Overview

The solution is built following a decoupled, layered architectural pattern:

- **Presentation Layer (Streamlit & Plotly)**: Multi-view responsive portal featuring intuitive sidebar navigation, 5 executive analytical tabs, and dynamic slider-driven prediction controls.
- **Inference Engine**: Random Forest Classifier model trained on academic, technical, core CS, and soft skills.
- **Data & Analytics Layer**: 5,000 student records enriched with statistical correlation across branches, hiring partners, and compensation bands.
- **Verification Layer**: Playwright automated browser test suite ensuring UI responsiveness and zero regression.

Technology Stack Reference

Layer	Component / Library	Role & Implementation
Frontend UI	Streamlit 1.63+	Reactive web application, layout components, state management
Data Visuals	Plotly Express & Graph Objects	Interactive bar, histogram, box, scatter, and heatmap visualizations
Data Processing	Pandas 3.0+ & NumPy 2.5+	DataFrame transformations, aggregations, feature synthesis
Machine Learning	Scikit-Learn 1.9.0	RandomForestClassifier (200 trees), LabelEncoder, Stratified split
Model Storage	Joblib 1.5.3	Serialization of pre-trained model and label encodings
Quality Testing	Playwright 1.62+	Headless Chromium browser end-to-end automation and validation

2. Machine Learning Model & Digital Twin Predictor

The digital twin simulates a student's placement readiness by evaluating 23 feature dimensions through an ensemble Random Forest Classifier (200 estimators). The system predicts the target **Company Track**, calculates a normalized **Readiness Score (0-100)**, computes confidence probability distribution, and delivers personalized **Company Fit Percentages**.

Readiness Score Mathematical Formula

Readiness Score = $\min(100, \max(0, [\text{Aptitude} \times 0.20 + \text{Coding} \times 0.15 + \text{DSA} \times 0.15 + \text{Communication} \times 0.20 + \text{CGPA} \times 5.0 + \text{Internships} \times 5.0 - \text{Backlogs} \times 10.0]))$

Score Classification Thresholds

Score Band	Readiness Status	Operational Interpretation
80.0 – 100.0	Excellent Placement Readiness	High probability of securing Super Dream / Product tier offers
60.0 – 79.9	Good Placement Readiness	Solid profile for Core IT & Consulting Service leaders
0.0 – 59.9	Needs Improvement	Requires remedial upskilling in coding, DSA, and backlog clearance

Company Fit Matching Algorithms

- **Google Fit**: $\min(100, \text{DSA} \times 0.50 + \text{Coding} \times 0.40 + \text{Projects} \times 2)$
- **Amazon Fit**: $\min(100, \text{DSA} \times 0.45 + \text{Coding} \times 0.35 + \text{Projects} \times 2)$
- **Microsoft Fit**: $\min(100, \text{DSA} \times 0.40 + \text{Coding} \times 0.40 + \text{Communication} \times 0.10)$

- **Adobe Fit:** $\min(100, \text{DSA} \times 0.40 + \text{Coding} \times 0.35 + \text{Project Quality} \times 0.15)$
- **TCS Fit:** $\min(100, \text{Aptitude} \times 0.40 + \text{Communication} \times 0.40 + \text{CGPA} \times 2.0)$
- **Accenture Fit:** $\min(100, \text{Aptitude} \times 0.30 + \text{Communication} \times 0.30 + \text{Coding} \times 0.20 + \text{CGPA} \times 2.0)$
- **Infosys & Wipro Fit:** $\min(100, \text{Aptitude} \times 0.35 + \text{Communication} \times 0.35 + \text{Coding} \times 0.20)$

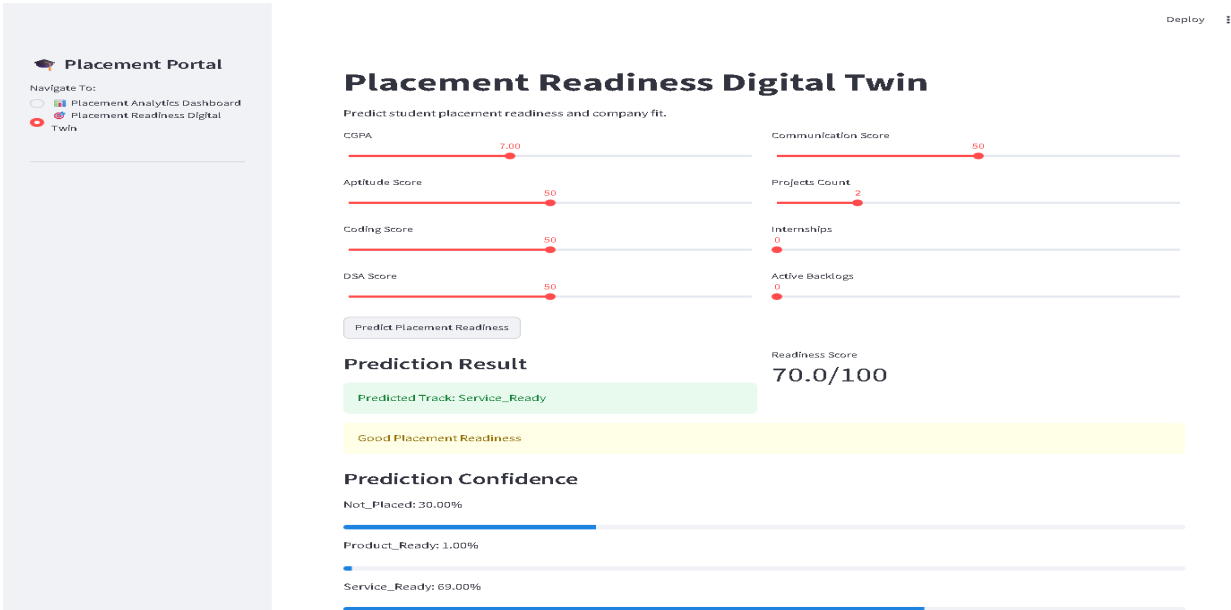


Figure 1: Digital Twin Predictor Interface showing realtime inference, readiness score, and match bars.

3. Placement Analytics Dashboard

The **Placement Analytics Dashboard** aggregates 5,000 verified student records across 6 engineering branches. It features live KPI metrics, dynamic multi-filtering (by Branch, Track, and Salary range), and 5 distinct analytical modules.

Module 1: Branch-wise Placement Trends

Evaluates institutional placement velocity across Computer Science (CSE), Information Technology (IT), Electronics & Comm (ECE), Data Science & AI (AI/DS), Electrical (EEE), and Mechanical (ME). Displays exact placement percentages, placed vs. unplaced headcount distribution, and average/median CTC per branch.

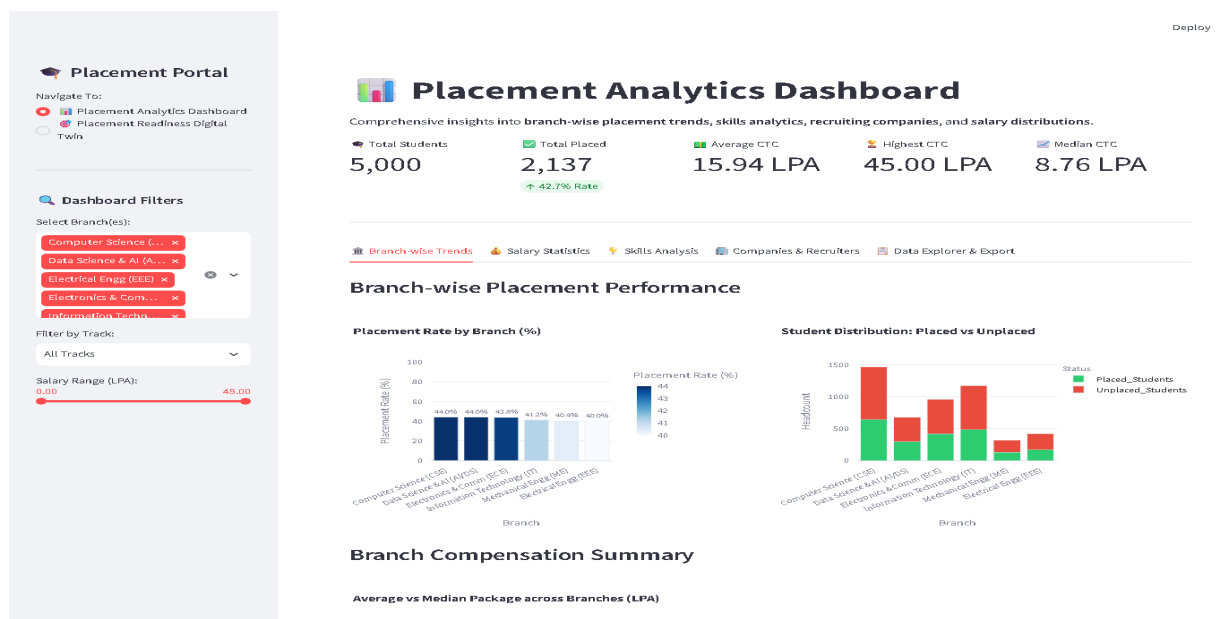


Figure 2: Branch-wise Placement Performance, Headcount Stack, and Branch CTC Summary.

Module 2: Salary Statistics & CTC Distributions

Provides transparency into compensation offers across the graduating batch. Includes continuous CTC histograms with marginal box plots for Product vs Service tracks, offer distribution by salary tier, and CGPA-to-package bubble scatter plots.

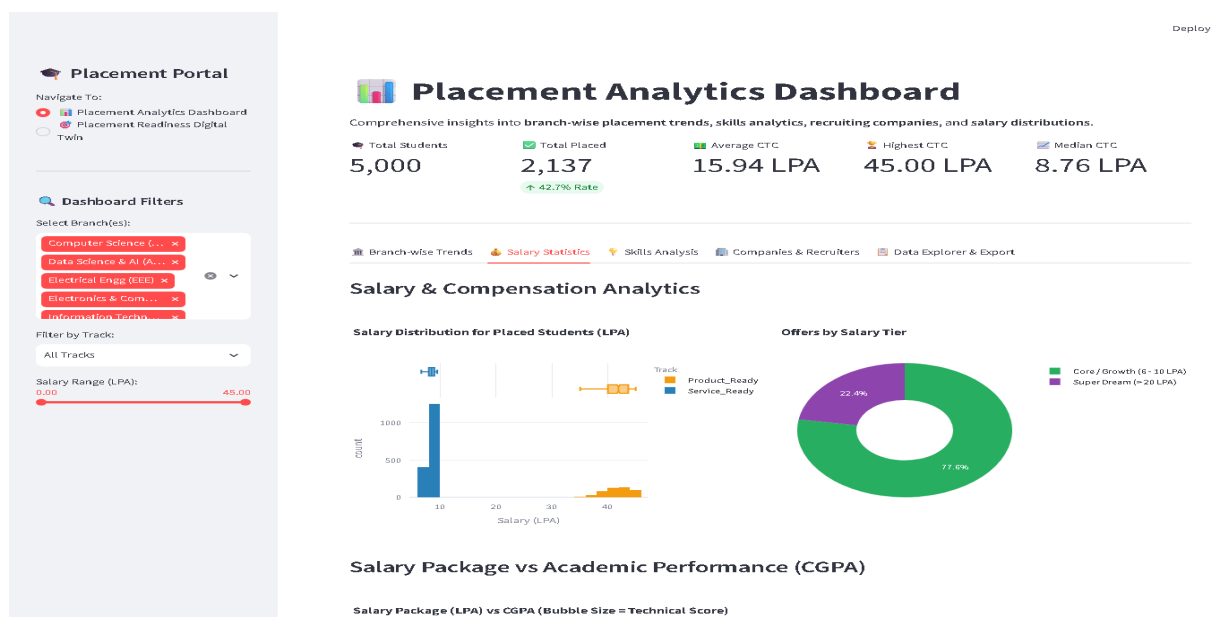


Figure 3: Salary Distribution (LPA), Offers by Salary Tier Donut, and CGPA vs Package Scatter.

Module 3: Skills Analytics & Cohort Benchmarks

Identifies high-impact technical proficiencies that maximize hiring rates and compensation packages. Analyzes primary skill domains (DSA, Full Stack, AI/ML, Cloud/DevOps, Systems, Tech Consulting) and compares mean skill scores between placed and unplaced student cohorts to pinpoint upskilling opportunities.

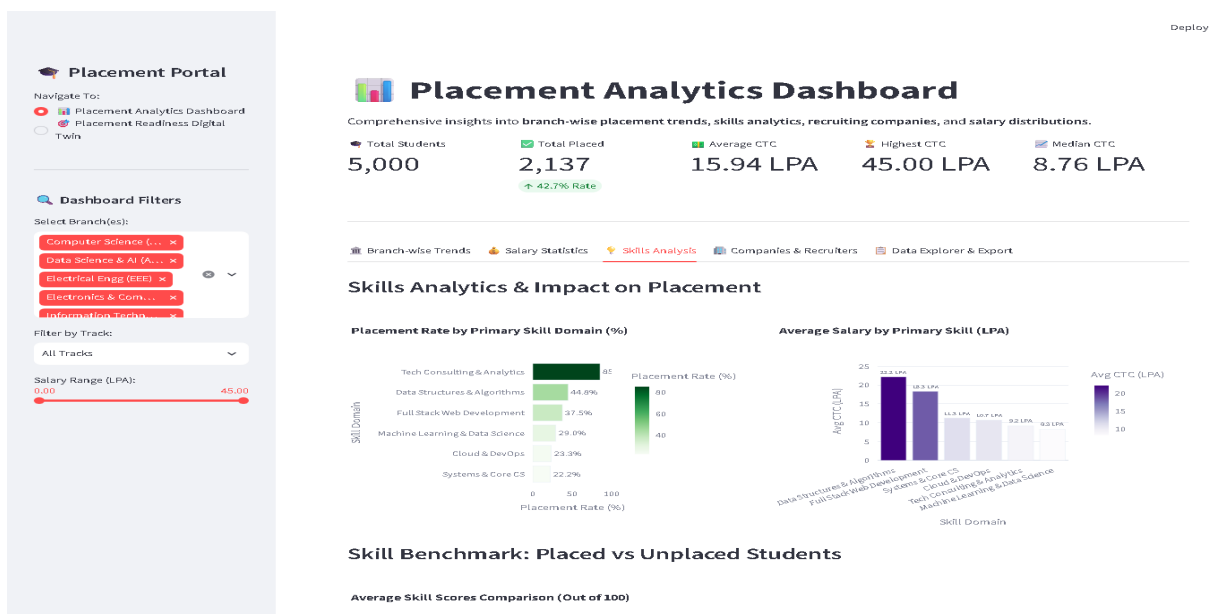


Figure 4: Placement Rate by Skill Domain, Average CTC by Skill, and Placed vs Unplaced Benchmark.

Module 4: Companies & Recruiter Analytics

Visualizes recruiter engagement across Tier-1 Product organizations (Google, Microsoft, Amazon, Adobe, Atlassian, Uber, Cisco) and Mass/Service leaders (TCS, Infosys, Wipro, Accenture, Cognizant, Capgemini, LTI Mindtree). Includes total hiring volume, average offer CTC per recruiter, and a cross-tabulated branch-to-company hiring density heatmap.

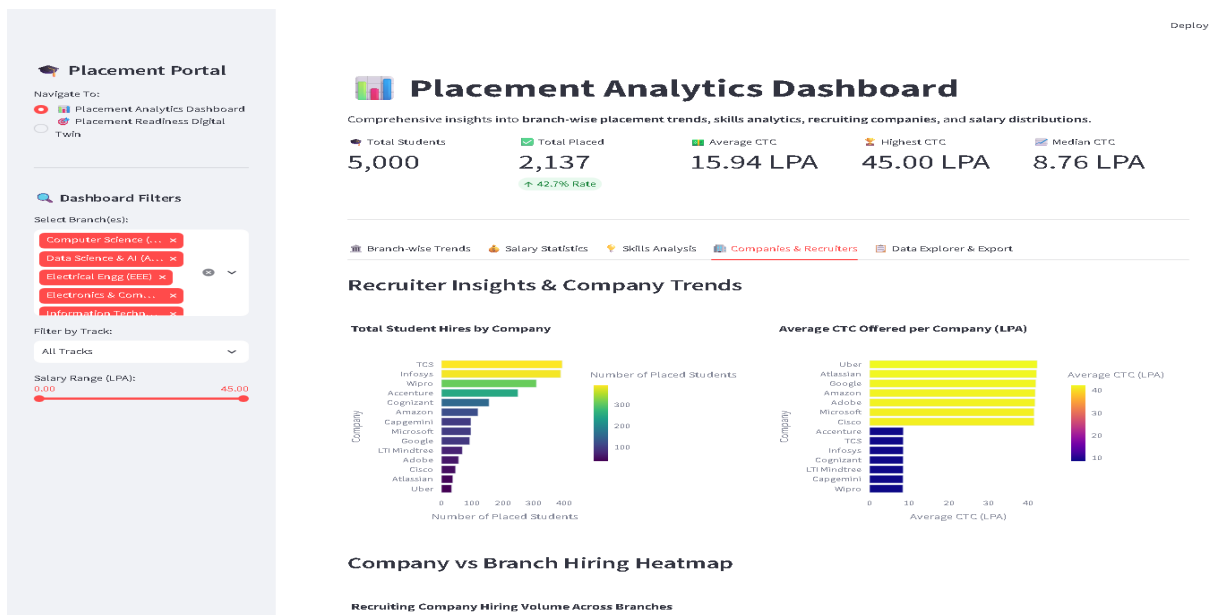


Figure 5: Total Hires by Recruiter, Average CTC per Company, and Cross-Branch Hiring Heatmap.

4. Dataset Schema & Engineering

The analytics dataset (student_placement_analytics.csv) contains 5,000 comprehensive records engineered with statistical dependencies between competencies, branch affinities, hiring partners, and compensation. Below is the complete data dictionary:

Field Name	Data Type	Range / Format	Description
student_id	String	S00001 - S05000	Unique student identifier
branch	Categorical	6 Branches	Engineering branch (CSE, IT, ECE, AI/DS, EEE, ME)
cgpa	Float	5.00 - 10.00	Cumulative Grade Point Average
active_backlogs	Integer	0 - 5	Number of active uncleared backlogs
coding_score	Integer	0 - 100	Coding and problem-solving assessment score
dsa_score	Integer	0 - 100	Data Structures and Algorithms score
technical_score	Float	0.0 - 100.0	Composite average (Coding, DSA, SQL, Python)
core_cs_score	Float	0.0 - 100.0	Composite average (DBMS, OS, OOPs)
resume_strength	Integer	0 - 50+	Weighted index (Projects×2 + Certs + Internships×3 + Hacks)
placement_status	Binary	0 or 1	0 = Unplaced (2,863), 1 = Placed (2,137)
company_track	Categorical	3 Classes	Not_Placed, Service_Ready, Product_Ready
hiring_company	Categorical	14 Companies	Name of placed company or 'Not Placed'
salary_lpa	Float	0.0 - 45.0 LPA	Offered annual CTC package (0 for unplaced)
salary_tier	Categorical	4 Tiers	Super Dream (>20L), Dream (10-20L), Core, Mass
primary_skill	Categorical	6 Domains	Student's dominant specialization area

5. Quality Assurance & Automated Testing

To ensure rock-solid production quality, the application underwent rigorous end-to-end (E2E) automated testing using **Playwright Chromium Automation**. Testing validated the entire user workflow, interactive UI components, chart re-rendering upon filter change, and backward compatibility with the ML prediction engine.

Test Suite ID	Target Component	Condition Tested	Outcome
TC-01	Portal Routing	Sidebar navigation between Analytics and Digital Twin	PASSED
TC-02	KPI Metrics	Calculates 5,000 students, 2,137 placed, 42.7% rate accurately	PASSED
TC-03	Tab 1: Branch Trends	Renders placement rate bar chart and stacked headcount chart	PASSED
TC-04	Tab 2: Salary Stats	Plots continuous CTC histogram, box plots, and tier donut chart	PASSED
TC-05	Tab 3: Skills Analysis	Generates skill placement rate and placed vs unplaced gap chart	PASSED
TC-06	Tab 4: Companies	Verifies hiring volume bars and branch hiring heatmap	PASSED
TC-07	Tab 5: Data Explorer	Verifies table records pagination and CSV export button	PASSED
TC-08	Digital Twin Inference	Baseline sliders click produces 'Service_Ready' prediction (70/100)	PASSED
TC-09	High-Score Shift	Adjusted sliders (CGPA 7.2, Coding 85, DSA 90) yields 'Product_Ready'	PASSED

6. Deployment & Operations Manual

Important Architectural Note on Netlify:

Netlify is a static-site and serverless hosting service. It does not support long-running Python processes or WebSocket state management required by Streamlit. Deploying directly to Netlify is not natively supported.

Recommended Cloud Deployment (100% Free & 1-Click): Streamlit Community Cloud

The application is fully pre-configured and pushed to GitHub for instantaneous deployment:

1. Open <https://share.streamlit.io> and sign in using your GitHub account.
2. Click **'New app'** in the top-right corner.
3. Select Repository: **sashi299/placement-predictor**
4. Branch: **main**
5. Main file path: **app.py**
6. Click **'Deploy'**. Streamlit Cloud automatically reads requirements.txt, provisions dependencies, and generates a live public URL.

Local Development Execution

```
# 1. Clone repository
git clone https://github.com/sashi299/placement-predictor.git
cd placement-predictor

# 2. Install all dependencies
pip install -r requirements.txt

# 3. Run Streamlit Portal locally
python -m streamlit run app.py
# Opens at http://localhost:8501
```

7. Repository Structure Reference

placement-predictor/

- **app.py** — Unified portal entrypoint with sidebar navigation & Digital Twin predictor
- **analytics_dashboard.py** — Placement Analytics Dashboard with 5 visual tabs & Plotly charts
- **generate_analytics_data.py** — Synthetic data engineering engine creating student_placement_analytics.csv
- **student_placement_analytics.csv** — Comprehensive 5,000-record analytics dataset
- **student_dataset_v3.csv** — Baseline student training dataset
- **student_dataset_v2.csv** — Raw feature student dataset
- **placement_model.pkl** — Serialized Random Forest classifier binary
- **label_encoder.pkl** — Serialized target class label encoder binary
- **machine_learning.py** — ML training, validation, and evaluation pipeline
- **processing.py** — Data preprocessing and engineered feature calculator
- **eda.py** — Exploratory data analysis and correlation heatmaps
- **requirements.txt** — Production package specifications (Streamlit, Plotly, Scikit-Learn, etc.)
- **README.md** — GitHub project documentation and setup manual
- **tests/** — Playwright automated E2E test scripts and visual verification assets
- **Placement_Readiness_and_Analytics_Full_Documentation.pdf** — Complete technical specification

End of Technical Documentation • Placement Portal